Aslin Garvasis

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	7.73
SHHSS Dwaraka	Class XII (Board of Higher Secondary Examinations, Kerala)	2023	99.67%
SJHSS Kallody	Class X (Kerala Board of Public Examinations)	2021	100.00%

SKILLS

Programming Languages: C, Python, MATLAB

Hardware Description Languages: Verilog HDL

EDA & Simulation Tools: LTspice, ModelSim, Quartus Prime Lite, GTKWave, Icarus Verilog

Version Control: GitHub github.com/Aslin2005

RELEVANT COURSES

Energy Management, Power Systems, Power Electronics, Electric Machines, Circuit and Network Analysis, Electronic Devices and Circuits, Semiconductor Device Fundamentals, Analog Electronics, Digital Systems, Control Systems, Signals and Systems, Digital Signal Processing, PCB Design and Manufacturing, Scientific Programming, Mathematics for Electrical Engineers, Vector Calculus, Differential Equations, Matrix Theory, Probability and Random Variables, Data Structures and Algorithms.

PROJECTS

Optical Lock-In Amplifier PCB *(Course Project - PCB Design and Manufacturing)*

- Designed and developed a 4-layer mixed-signal PCB for an STM32H743-based optical lock-in amplifier capable of detecting nanovolt-level signals (0.1 Hz-50 kHz).
- Designed a precision analog front-end using the INA826 instrumentation amplifier with programmable gain and a low-noise dual-regulator power architecture.
- Completed schematic design, PCB layout, controlled-impedance routing, and Gerber generation using Altium Designer.

Common-Source MOSFET Amplifier *(Course Project - Electronic Devices and Circuits Lab)*

- Designed and implemented a Common-Source MOSFET amplifier using a 2N7000 MOSFET with a target midband voltage gain of 50.
- Performed small-signal analysis and experimentally characterized the frequency response from 10 Hz to 10 MHz.
- Validated the design by comparing measured gain and phase response with theoretical and simulated results.

Embedded Scientific Calculator *(Course Project - Scientific Programming)*

- Developed a scientific calculator using an AVR microcontroller with LCD and keypad interface.
- Implemented arithmetic and scientific functions using Embedded C.
- Integrated expression parsing, numerical computations, and peripheral interfacing.

Pipelined Radix-2 8-Point FFT Processor *(Course Project - DSP Laboratory)*

- Designed and implemented a pipelined Radix-2 8-point FFT processor with streaming input architecture using Verilog HDL.
- Verified functional correctness against MATLAB implementation(for verilog codes) and Intel FFT IP Core

outputs(Quartus Prime Lite).
- Evaluated resource utilization, throughput, and maximum operating frequency (Fmax) using Quartus Prime Lite and optimized the design for improved performance.

Sallen-Key Active Filters *(Course Project - Electronic Devices and Circuits Lab)*

- Designed, simulated, and implemented active low-pass and band-pass filters using LM358 operational amplifiers.
- Analyzed frequency response and compared experimental results with theoretical calculations.
- Validated circuit performance through hardware implementation and testing.

Mess Entry Counter *(Course Project - Electronic Devices and Circuits Lab)*

- Designed and implemented a synchronous up/down counter using JK flip-flops and TTL logic ICs.
- Interfaced a 7447 BCD-to-seven-segment decoder for real-time count display.
- Verified functionality through hardware implementation and testing.

POSITIONS OF RESPONSIBILITY

Events & PR Coordinator, SPIC MACAY, IIT Hyderabad

- Planned and coordinated institute-level cultural events as part of the SPIC MACAY chapter.
- Led publicity and outreach efforts through online and offline communication to increase event participation.
- Coordinated with artists, faculty, and student volunteers to ensure seamless event execution.

EXTRACURRICULAR ACTIVITIES

Serving as an active NSS (National Service Scheme) volunteer contributing to community development initiatives.